

Chapter-5

STRUCTURED KNOWLEDGE REPRESENTATION

5.1 Introduction

Structural knowledge is basic knowledge to problemsolving. It describes relationships between various concepts such as kind of, part of, and grouping of something. It describes the relationship that exists between subjects to concepts or objects.

5.2 Knowledge Model

A *model* is a world in which a sentence is true under a particular interpretation. There can be several models at once that have the same interpretation. First order predicate logic consists of objects, predicates on objects, connectives, and quantifiers. *Logic* is simple representation which helps to infer new fact from the existing information. *Predicates* are the relations between objects or properties of the objects. *Connectives* and *quantifiers* allow for universal sentences. Relation between objects can be true or false. E.g., propositional logic.

We had already discussed logic representation in detail in chapter 4. Here in this chapter we focus much in structured knowledge representation namely semantic network, frame and scripts.

5.3 Semantic Network

Semantic network is an alternative to predicate logic as a form of knowledge representation. Semantic network is a declarative graphic representation that can be used to represent knowledge and support automated systems for reasoning about the knowledge. The structure of a semantic net is shown graphically in terms of nodes and the arcs connecting each other. Nodes are sometimes referred to as objects, events, subjects while arcs

represent the links or relations. The links are used to express relationships.

It is argued that this form of representation is closer to the way human's structure the knowledge as human also store the knowledge with relational link. For example; if a man lost bicycle key then he remembers where he had been, then he had used his bicycle last time etc, making this kind of relating scenario he goes to search those places.

Advantages of semantic web/network:

- This method is easy to visualize.
- It is efficient in space requirement.
- The objects are represented only once.

Disadvantages of semantic web/network:

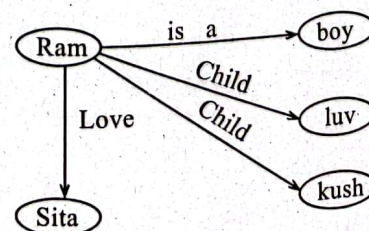
- It is unable to represent negation, quantification, disjunction etc.
- We cannot infer or deduct new information.

Example 5.1:

Show the following statements in semantic network.

1. Ram is a boy.
2. Ram loves Sita.
3. Ram's children are luv and kush.

Solution:



Example 5.2:

Show the following statements in semantic network

1. Tom is a cat.
2. Tom caught a bird.
3. Tom is owned by John.
4. Tom is ginger in color.
5. Cats like cream.
6. The cat sat on the mat.
7. A cat is a mammal.
8. A bird is an animal.
9. All mammals are animals.
10. Mammals have fur.

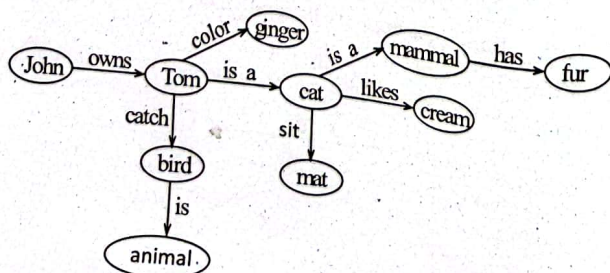
Solution:

We can make individual diagram with ease.

Tom is a cat (Tom) is a (Cat)

Tom caught a bird (Tom) catch (Bird)

Representing all in a single picture we get the following diagram.

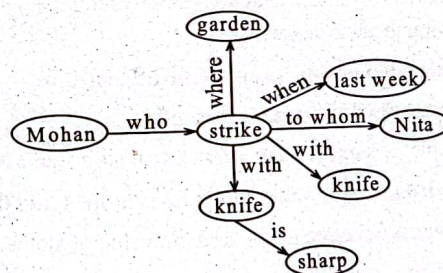


Example 5.3:

Show the following statement in semantic network.

Mohan struck Nita in the garden with a sharp knife last week.

Solution:



5.4. Frames

Frame is a static data structure used to represent well understood situation in a group of slots and slot fillers. Frame is similar to a record structure. It is used in many AI applications including vision and natural language processing that provides a convenient structure for representing. A single frame is not much useful. Frame systems usually have collection of frames connected to each other. Frames are also useful for representing commonsense knowledge. While semantic nets are basically a two-dimensional representation of knowledge, frames add a third dimension by allowing nodes to have structures. By using frame structure, we can use filler slots and inheritance, very powerful knowledge representation systems can be built. Frame-based expert systems are very useful for representing causal knowledge because their information is organized by cause and effect. Frames are generally designed to represent either generic or specific knowledge.

Frame structure contains following information:

- **Frame identification name**

It is field written in top of the frame structure where name of the frame is placed.

Example: A frame which stores knowledge about a car can have frame name as car.

- **Relationship of this frame to the other frame**

It relates different frame to each other.

Example: A superclass of a frame (car) is a frame (vehicle).

- **Knowledge about an attribute of an object and its value**

Attribute are written in slot and the value of slot is written in slot filler.

Example: A frame (car) can have an attribute as no. of wheels with value 4.

- **Frame default information**

These are slots values that are taken to be true when no evidence to the contrary of frame has been formed.

Example: 'Ram's car' frame copies the slot and slot values of its parent frame 'car' like no of wheel, model name etc.

5.4.1 Types of Frame

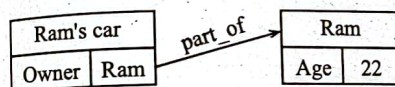
1. **Class frame:** It is the main frame from which other frames can be inherited.
2. **Subclass frame:** It is frame inherited from class frame.
3. **Instance frame:** This frame is bottom frame from which no other frame can be derived. This frame may be derived from both subclass and class frame.

5.4.2 Types of Relationship

1. **Is-a relationship:** It relates subclass frame with a class frame or an instance frame with a subclass or class frame. In

this case a subclass frame or instance frame inherits all slots from a class frame and it can also include new slots.

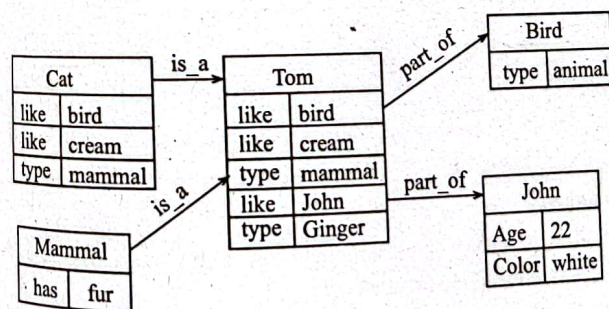
2. **Part of relationship:** It relates the slot values with its constituent parts:



3. **Semantic relationship:**

It relates object with its attributes and frame structure can be represented in semantic network.

Solution for the same example in frame can be shown in frame structure as shown below:

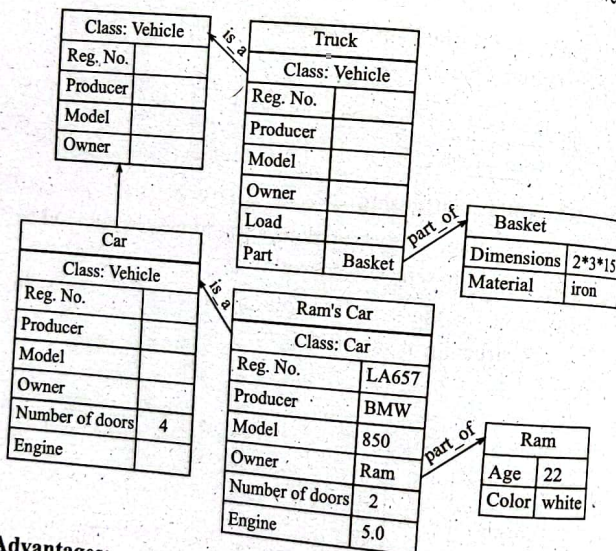


5.4.3 Frame Methods

Methods in frame are also called *demons* which are attached to slots. Demons are automatically invoked when a slot is accessed. Standard demons are:

1. **IF NEEDED:** It is invoked when it is necessary to acquire a slot value.
2. **IF CHANGED:** It is invoked when a value of slot is changed.
3. **IF ADDED:** It is invoked when a value is added to a slot.
4. **IF REMOVED:** It is invoked when value of slot is deleted.

Example below shows relations between relations between class, subclass, instance frame.



Advantages:

1. It makes programming easier by grouping related knowledge together.
2. Easy to setup new slots or new properties and relations.
3. Easy to include default information and detect missing values.

Disadvantages:

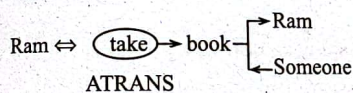
1. Frame for a room will be completely different person.
2. No associated inference mechanism.

5.5 Conceptual Dependency and Script

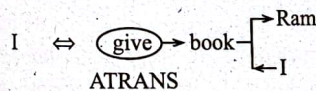
Conceptual dependency is a theory in which we try to be independent of the words. For any two or more sentences that are identical in meaning, there should be only one representation. A

script is a remembered precedent, consisting of tightly coupled, expectation-suggesting primitive-action and state change frames. A script is a structured representation describing a stereotyped sequence of events in a particular context i.e., extend frames by explicitly representing expectations of actions and state changes. Find primitives to describe the world like PTRANS for "transfer physical location of an object (= go)" and ATRANS for "transfer a relationship (= give)".

E.g., Ram took a book from someone.



I give a book to Ram.



Conceptual dependency provides:

- a structure into which nodes representing information can be placed
- a specific set of primitives
- at a given level of granularity.
- Sentences are represented as a series of diagrams depicting actions using both abstract and real physical situations.
- The agent and the objects are represented.
- The actions are built up from a set of primitive acts which can be modified by tense.

Examples of primitive acts are:

- ATRANS - Transfer of an abstract relationship. e.g. give, take, lend, borrow.
- PTRANS - Transfer of the physical location of an object. e.g. go, went.

- PROPEL - Application of a physical force to an object. e.g. push, pull.
- MTRANS - Transfer of mental information. e.g. tell.
- MBUILD - Construct new information from old. e.g. decide, conclude.
- SPEAK - Utter a sound. e.g. say, talk, play music.
- ATTEND - Focus a sense on a stimulus. e.g. listen, watch.
- MOVE - Movement of a body part by owner. e.g. punch, kick.
- GRASP - Actor grasping an object. e.g. clutch, collect.
- INGEST - Actor ingesting an object. e.g. eat, drink.
- EXPEL - Actor getting rid of an object from body. e.g. throw, sweat, cry, excrete.

Script is originally developed to represent knowledge acquired from natural language input. To implement script, we use conceptual dependency.

A script is composed of:

1. Entry condition must be true for further execution of the script.
2. Results or facts are true once the script is executed.
3. Props or things that make up the content of the script.
4. Roles are the actions that the individual participant performs.
5. Scene which represent temporal aspect of script.
6. Tracks are the variation on script. Different track may share different component of same script.

E.g.,

Entry condition:

1. Man is Hungry.
2. Restaurant should be open.

Result:

1. Man has less money.
2. Man is less hungry.

Props:

1. Table (T)
2. Menu (ME)
3. Glass (G)
4. Door (D)
5. Chair (C)
6. Table (T)
7. Restaurant (R)

Roles:

1. Customer (M)
2. Waiter (W)
3. Chef (Ch)

Scene 1: Get to table

M PTRANS to D
M PROPEL D
M ATTEND T
M MBUILD For T

Scene 2: Eat the food

Track 1: for menu in the table

M ATTEND ME
M MBUILD F
M SPEAK to W
M ATRANS F from W
M INGEST F

Track 2: for menu not in the table

M SPEAK with W for ME
M ATRANS ME from W
M ATTEND ME
M MBUILD F

M SPEAK to W
M ATRANS F from W
M INGEST F

Scene 3: Exit from the restaurant

M MOVE
M PTRANS.to D
M PROPEL D
M PTRANS from R

EXERCISE

1. Elaborate the concept of semantic network.
2. What is frame? Explain its type, class, methods.
3. How are scripts written? Illustrate it with example.
4. What is the importance of the conceptual dependency in artificial intelligence?

Chapter 6

MACHINE LEARNING

6.1 Introduction

Learning is the process of acquiring new and better or updated information from existing knowledge, behaviors, skills, values, or preferences. The ability to learn is possessed by humans, animals, and some machines; there is also evidence for some kind of learning in some plants. Some learning is immediate, induced by a single event (e.g., being burned by a hot stove), but much skill and knowledge accumulates from repeated experiences. The changes induced by learning often last a lifetime, and it is hard to distinguish learned material that seems to be "lost" from that which cannot be retrieved.

Learning is the ability of an agent to improve its behavior based on experience. This could mean the following:

- The range of behaviors is expanded; the agent can do more.
- The accuracy on tasks is improved; the agent can do things better.
- The speed is improved; the agent can do things faster.

The ability to learn is essential to any intelligent agent. Learning involves an agent remembering its past in a way that is useful for its future. This chapter considers supervised learning i.e., given a set of training examples made up of input-output pairs, predict the output of a new input.

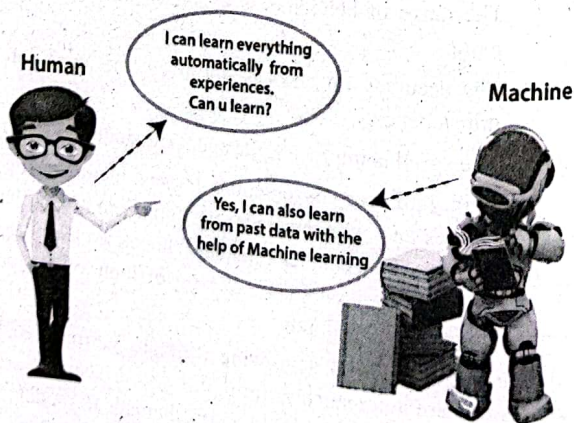
Machine learning is a growing technology which enables computers to learn automatically from past data. Machine learning uses various algorithms for building mathematical models and making predictions using historical data or information. Currently, it is being used for various tasks such as image recognition, speech recognition, email filtering, Facebook auto-tagging, recommender

system, and many more. Machine learning is further divided into *supervised*, *unsupervised*, and *reinforcement learning*.

In the real world, we are surrounded by humans who can learn everything from their experiences with their learning capability, and we have computers or machines which work on our instructions. But can a machine also learn from experiences or past data like a human does? So here comes the role of Machine Learning.

Machine learning is said as a subset of artificial intelligence that is mainly concerned with the development of algorithms which allow a computer to learn from the data and past experiences on their own. The term machine learning was first introduced by Arthur Samuel in 1959. We can define it in a summarized way as:

Machine learning enables a machine to automatically learn from data, improve performance from experiences, and predict things without being explicitly programmed.



With the help of sample historical data, which is known as training data, machine learning algorithms build a mathematical model that helps in making predictions or decisions without being explicitly programmed. Machine learning brings computer science

and statistics together for creating predictive models. Machine learning constructs or uses the algorithms that learn from historical data. The more we will provide the information, the higher will be the performance.

A machine has the ability to learn if it can improve its performance by gaining more data.

How does Machine Learning Work?

A machine learning system learns from historical data, builds the prediction models, and whenever it receives new data, predicts the output for it. The accuracy of predicted output depends upon the amount of data, as the huge amount of data helps to build a better model which predicts the output more accurately. Suppose we have a complex problem, where we need to perform some predictions, so instead of writing a code for it, we just need to feed the data to generic algorithms, and with the help of these algorithms, machine builds the logic as per the data and predict the output. Machine learning has changed our way of thinking about the problem. The block diagram below explains the working of machine learning algorithm:

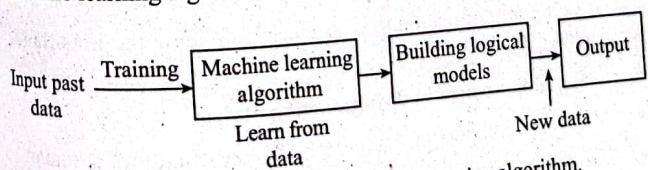


Figure 6.1: Steps involved in machine learning algorithm.

Features of machine learning:

- Machine learning uses data to detect various patterns in a given dataset.
- It can learn from past data and improve automatically.
- It is a data-driven technology.
- Machine learning is much similar to data mining as it also deals with the huge amount of the data